

1. INPUT SPECIFICATIONS

Required capacity	100.0 t/h
Lift height	25.0 m
Material	Clinker
Bulk density	1500 kg/m3
Service environment	Dry
Site wind pressure	800 Pa

2. PERFORMANCE SUMMARY

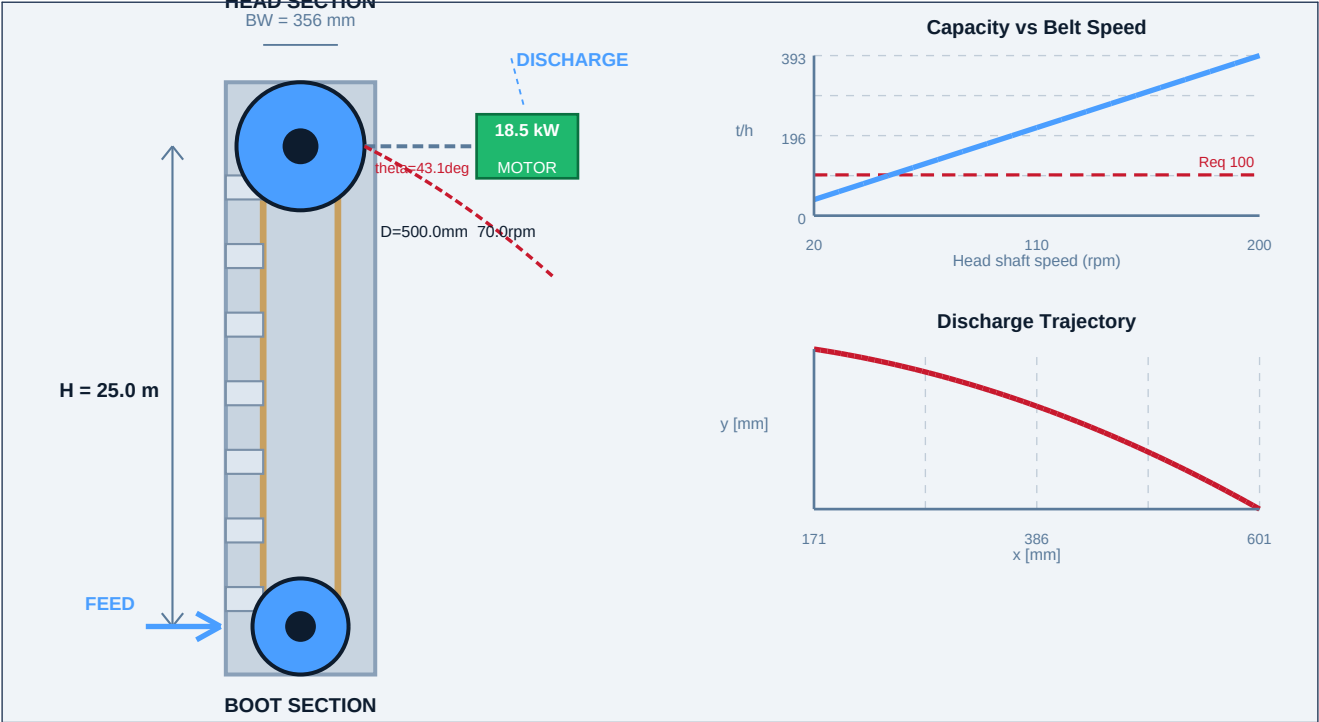
Capacity 137.4 t/h	Belt Speed 1.83 m/s	Total Power 13.0 kW	Motor Selected 18.5 kW	Chain SF 20.85 —	Bearing L10 129,166 h
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3. PROCESS LEVEL OUTPUTS

Design Configuration

Head pulley dia.	500.0 mm	Belt type	EP
Head shaft speed	70.0 rpm	Belt friction mu	—
Boot pulley dia.	300.0 mm	Wrap angle	— deg
Fill factor	45%	Service factor	—

System Schematic & Performance Charts



Achieved capacity	137.4 t/h	Discharge angle	43.1 deg from vertical
Chain speed	1.83 m/s	Belt width	356 mm
Bucket spacing	0.278 m	Recommended fill	45.0%

Centrifugal ratio	1.369	Stream spread	0.124
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4. POWER BREAKDOWN

Boot equivalent height (D_boot x Leq)	4.00 m
Total equivalent height (H + H_equiv)	29.00 m
Lift power (material only)	9.36 kW
Boot scooping power (boot loading)	1.50 kW
Shaft power (total load at head shaft)	10.86 kW
Drive losses (gearbox + bearings + belt flex)	2.17 kW
Total motor power (shaft x Ceff)	13.03 kW
Design power (x service factor 1.25)	16.29 kW
Selected motor	18.5 kW
Gearbox ratio (at 1450 rpm input)	20.7 : 1
Drive efficiency factor Ceff	1.200
CEMA LEQ factor	8.0

5. COMPONENT DESIGN OUTPUTS

5a. Head Shaft & Drive

Material grade	ASTM A36 Mild Steel	Hub connection	Keyed (ASME B17.1)
Shaft torque	1778.1 Nm	Hub OD	166.2 mm
Section	Solid	Hub length	138.5 mm
Min shaft dia.	110.8 mm	Key b x h	28 x 16 mm
Governed by	stress	Key shear	8.28 / 42.0 MPa
Allowable shear	42 MPa (keyed)	Key bearing	28.97 / 215.0 MPa
Bearing span	686 mm	Key result	PASS
Drive arm A	309 mm	Bearing L10	129166 h
Tail arm B	377 mm		

5b. Belt & Bucket Selection

Bucket series	AC_12x8	Belt width	356 mm
Style	CC	Belt class	1 PLY
Width x depth	305 x 216 mm	Belt length (total)	53.1 m (incl. splice allowance)
Projection	203 mm	Bucket spacing (actual)	0.277 m
Volume (struck)	8.58 L	Bucket count	186 off
Bucket mass	11.00 kg	Fill factor	45%
		Material DB fill	45.0% advisory

5c. Pulley Design

Lagging type	—	Shell min t (CEMA)	11.0 mm
Lagging thickness	— mm	Shell min t (press)	0.67 mm
mu dry / wet	— / —	Shell governing t	11.0 mm (CEMA minimum)
mu operating	—	Shell specified t	11.0 mm (= calculated minimum, no override)
Euler limit	—	End disc min t	22.8 mm
Belt ratio R/T3	—	End disc specify	27 mm (+20%)
Slip check	—	Disc governed by	plate_bending

		Disc sigma bend	80.0 MPa
		Force per disc	21.76 kN
		Shaft critical speed	2147 rpm
		Operating ratio	70 rpm (3% of critical)

5d. Take-Up Design

Type	Gravity (primary)	Alt type	Screw (alternative)
Counterweight net	3654.0 kg	Screw load	42.17 kN
Counterweight gross	4019.4 kg	Min core dia.	29.9 mm
Travel required	474 mm	Turns required	79.0
— thermal	48 mm	Buckling SF	5.11
— elongation	126 mm	Buckling check	PASS

5e. Casing & Structural

Plate thickness	8.2 mm	Panel delta actual	2.99 mm
Max stiffener pitch	2065.0 mm	Panel delta allowed	4.88 mm
Recommended pitch	1755.0 mm	Panel sigma max	11.1 MPa
Deflection limit	L / 360	Panel check	PASS
Wind pressure	800.0 Pa	Bolt fatigue Goodman	0.003
		Assembly fasteners	M12 x 793 (panel-to-panel + stiffener)

5f. Discharge Chute Design

Back-plate angle	60.2 deg	Spout width	405.0 mm
Min angle (wall fric.)	40.0 deg	Throw distance	0.144 m
Angle adequate	YES	Throat velocity	0.912 m/s
Flow regime	MASS_FLOW	Geom plugging risk	LOW
Mass-flow threshold	31.5 deg	Hood radius	0.312 m
Governed by	angle of repose	Capture efficiency	0.83
Wear index	16.79 (HIGH)	Plugging risk	HIGH
Liner material	AR500 Wear Plate	Plugging index	1.205
Liner thickness	12 mm	Dust risk	MEDIUM
Liner grade	500 BHN	Recommended sensors	liner_wear_indicator , flow_sensor_acoustic, chute_blockage_detector

Chute Recommendations

* Plugging index 1.21 (HIGH) — Air cannons or fluidising pads; steep chute angle required.

* Wear index 16.8 (HIGH) — specify AR500 Wear Plate (500 BHN), 12 mm thick.

5g. Feed Design (Boot Section)

Loading method	Centrifugal – Digging / Scooping	Material depth	152.0 mm (0.75xP)
Volumetric flow	91.63 m3/h	Dig zone arc	393.0 mm
Inlet velocity	0.3 m/s	Dig volume	21.3 L
Inlet area req.	848.4 cm2	Surge volume	76.4 L (3.0s)
Inlet width	356.0 mm	Boot casing h	719.0 mm
Inlet height	152.0 mm	Floor clearance	50.0 mm

5h. Chain Drive Configuration

Chain series	6102-1/2 (12" SC double)	Chain pull	25.60 kN
Chain pitch	304.8 mm	SF actual	20.85
No. of strands	2	Speed check	FAIL
Working load	27215 kg / strand	Sprocket teeth	10
Chain weight	11.0 kg/m	Sprocket PD	986.4 mm
Max chain speed	0.51 m/s	Smooth op.	YES

6. ENGINEERING VERIFICATION

Check	Actual	Limit	SF	Result
Capacity	137.4 t/h	100.0 t/h	1.37	✓ OK
Chain working load SF	20.85	>= 6.0 (CEMA 375)	20.85	✓ OK
Chain speed $v \leq$ rated	1.83 m/s	0.51 m/s rated	0.28	✗ FAIL
Sprocket 10–20 teeth (smooth)	10 teeth	10 – 20	—	✓ OK
Headshaft radial load	43.53 kN	80 kN	1.84	✓ OK
Shaft diameter (stress)	110.8 mm calc	110.8 mm governing	1.00	i INFO
Bearing L10 life	129,166 h	20,000 h min	6.46	✓ OK
Keyway shear stress	8.28 MPa	42.0 MPa	5.07	✓ OK
Keyway bearing stress	28.97 MPa	215.0 MPa	7.42	✓ OK
Bolt fatigue Goodman ratio	0.003	< 1.000 (infinite life)	333.33	✓ OK
Casing panel deflection	2.99 mm	4.88 mm (L/360)	1.63	✓ OK
Casing clearance (stream vs wall)	601.4 mm stream reach	228.0 mm wall	-373.4 mm margin	✗ FAIL
Stream → chute interception	39.4 deg impact angle	2.09 m/s impact velocity	—	✓ OK

Corrective Actions

■ PLUGGING — Chute plugging probability HIGH despite adequate angle — cohesive material

1. Specify AR500 Wear Plate liner to reduce surface adhesion
2. Install vibration motor or air cannon on chute back-plate
3. Consider chute geometry with steeper secondary slope

7. DESIGN NOTES

Capacity & Selection: The elevator is configured with Series AC_12x8 buckets and achieves a throughput of 137.4 t/h against the 100 t/h requirement, providing a capacity margin of 37%. Belt speed is 1.83 m/s at the specified head shaft speed, which is within the acceptable range for the selected bucket geometry. The centrifugal discharge ratio of 1.369 is within the optimal range for clean centrifugal discharge.

Head Shaft & Hub: The head shaft minimum diameter is 111 mm, governed by stress. The shaft spans 686.0 mm between bearing centrelines. The hub outer diameter is 166.2 mm with a hub engagement length of 138.5 mm. The keyway stress check passes at the selected shaft and hub dimensions.

Pulley Lagging: Rubber herringbone lagging, — mm thick, is specified for the head pulley. This provides a belt-to-pulley traction coefficient of — under dry service conditions. Belt slip analysis confirms the lagging selection is adequate at operating tensions.

Pulley End Disc: Pulley end discs require a structural minimum thickness of 22.8 mm (governed by plate_bending). A specified thickness of 27 mm is recommended in fabrication drawings to provide adequate construction tolerance and weld preparation allowance. Full finite element analysis or detailed annular plate calculation is recommended prior to fabrication.

Take-Up System: The gravity take-up counterweight gross mass is 4019.4 kg. The take-up frame must accommodate a minimum carriage travel of 474 mm, combining thermal expansion, belt elastic elongation, and installation clearance components. A 20% allowance above this minimum is recommended for field adjustment headroom. As an alternative for short elevators, a screw take-up requires a minimum screw core diameter of 29.9 mm. The screw buckling safety factor is adequate.

Casing & Structural: The casing plate thickness of 8.2 mm is determined by material bulk density and elevator height. Structural stiffeners at 1755.0 mm centres limit panel deflection to within acceptable serviceability limits under the 800.0 Pa design wind load. Panel deflection analysis confirms the stiffener layout is adequate.

Power & Motor: The calculated shaft power is 10.861 kW. Total motor input power including drive losses is 13.0 kW. A 18.5 kW motor has been selected, representing a 42% power margin above the design requirement at the specified service factor. The gearbox reduction ratio should be confirmed with the selected motor supplier based on motor synchronous speed.

Bearing Life: Head shaft bearing life is calculated at 129,166 hours, which exceeds standard requirements. Bearing selection should be confirmed from the manufacturer catalogue using the equivalent dynamic load at the specified shaft speed. Lubrication interval and replacement schedule should be established based on the supplier's rated life data for the selected bearing series.

Bucket Bolt Fatigue: Bucket mounting bolt fatigue has been assessed using the Goodman diagram method. The fatigue utilization ratio is 0.003, which is well within the infinite life threshold — bolt fatigue is not a design concern at the specified operating conditions.

Discharge Chute: The discharge chute back-plate angle of 60.2 degrees is set by the material trajectory. Flow analysis indicates mass flow conditions at the specified belt speed. AR500 Wear Plate liner, 12 mm thick, is specified based on the calculated wear index. Plugging probability is high and dust generation risk is medium for this material. The material throw distance is 0.144 m from the pulley centreline, which should be verified against the physical head section geometry.

Feed Design (Boot Section): The elevator uses centrifugal digging to fill buckets at the boot. The material depth in the boot pit should be maintained at approximately 152.0 mm (0.75 times the bucket projection) for consistent scooping. The active digging zone spans 393.0 mm of arc at the boot pulley, with an estimated digging volume of 21.3 litres. The boot surge volume is 76.4 L (3.0-second buffer). The boot casing floor should be a minimum of 719.0 mm below the boot pulley centreline to provide adequate clearance.

8. ENGINEERING SIGN-OFF

This report has been prepared in accordance with CEMA No. 375-2017 and ANSI/CEMA 550-2020. The signatory confirms that the design calculations have been reviewed and are accepted as the basis for procurement and fabrication.

DESIGNED BY Name: _____ Designation / Company: _____ Date: _____ Signature _____ STAMP / SEAL	REVIEWED BY Name: _____ Designation / Company: _____ Date: _____ Signature _____ STAMP / SEAL	APPROVED BY Name: _____ Designation / Company: _____ Date: _____ Signature _____ STAMP / SEAL
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Standards: CEMA No. 375-2017 · ANSI/CEMA 550-2020 · ISO 281 · ASME B17.1 · AS 4024 (where applicable)